

PROJECT 06

File Server and Departmental Permissions

Windows Server 2025 | SMB | NTFS | AGDLP | Access-Based Enumeration

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ENVIRONMENT	Personal Windows Enterprise Lab
VERSION	v1.0.0
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STATUS	Complete

Executive Summary

Configured **CORP-SRV01** as a departmental file server for the **corp.mslab.test** domain. The project added a dedicated NTFS data volume, created a centralized SMB share, implemented department-specific permissions through the existing AGDLP groups, enabled access-based enumeration, and validated both authorized modification and cross-department isolation from MGMT-CL01.

Business Problem

Departmental data requires centralized storage without exposing Finance, Human Resources, Information Technology, or Operations content to unrelated users. Direct user permissions are difficult to maintain and audit. This implementation completes the resource-permission layer of the AGDLP design introduced in Project 04.

Project Objectives

- Add a dedicated data disk to CORP-SRV01.
- Format a healthy NTFS volume labeled CorporateData.
- Create four departmental folders beneath a common share root.
- Apply permissions through domain-local resource groups in the GUI.
- Publish an SMB share with access-based enumeration and no offline caching.
- Validate share configuration and effective access with PowerShell.
- Prove authorized write/modify access and unauthorized cross-department isolation.

RESULT

The representative msalazar test could see, create, and modify IT content while Finance remained concealed and inaccessible.

Authorization and Storage Design

Layer	Implementation
Storage	Dedicated 60-GB VHDX attached to CORP-SRV01
Volume	E: NTFS CorporateData
Share root	E:\Shares
UNC path	\\CORP-SRV01\Departments
Enumeration	AccessBased
Offline caching	Disabled

AGDLP Permission Path

Department	Account group	Resource group	NTFS right
Finance	GG_Finance_Users	DL_FS_Finance_RW	Modify
Human Resources	GG_HR_Users	DL_FS_HR_RW	Modify
Information Technology	GG_IT_Users	DL_FS_IT_RW	Modify
Operations	GG_Operations_Users	DL_FS_Operations_RW	Modify

Share permissions allow Domain Users to connect with Change access. NTFS permissions remain authoritative at each departmental folder. Effective access is the most restrictive combination of share and NTFS permissions.

MODEL

Account -> Global group -> Domain Local group -> NTFS permission

Dedicated Data Disk

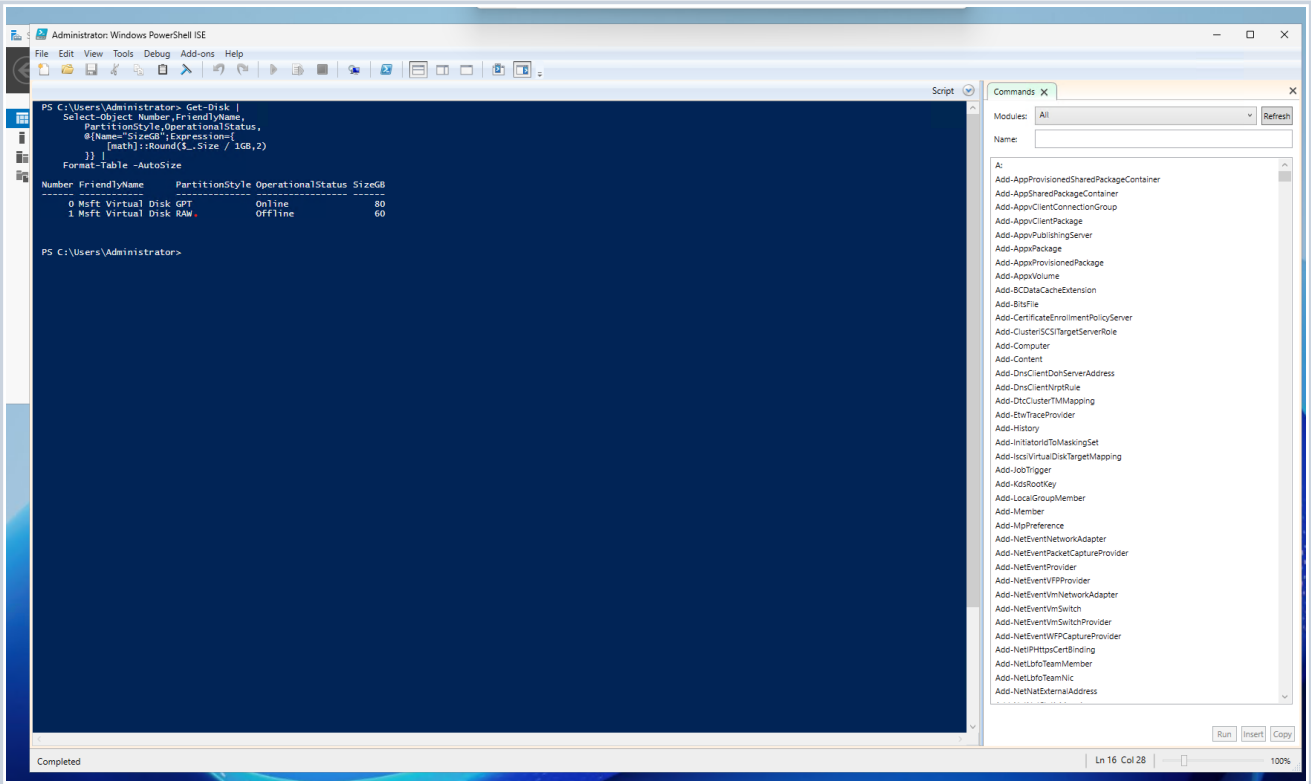


Figure 1 - New virtual disk discovered
PowerShell identifies Disk 1 as a 60-GB RAW, offline Microsoft virtual disk before initialization.

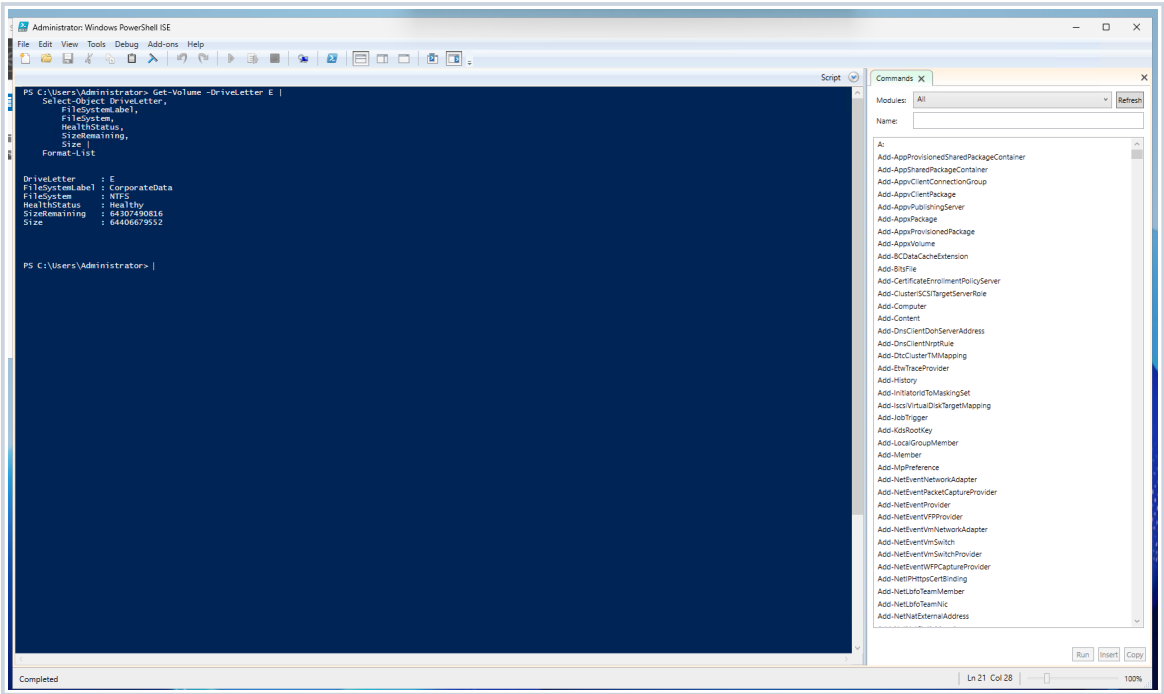


Figure 2 - CorporateData volume validation
The resulting E: volume is NTFS, Healthy, labeled CorporateData, and provides approximately 60 GiB of capacity.

Departmental Folder Structure

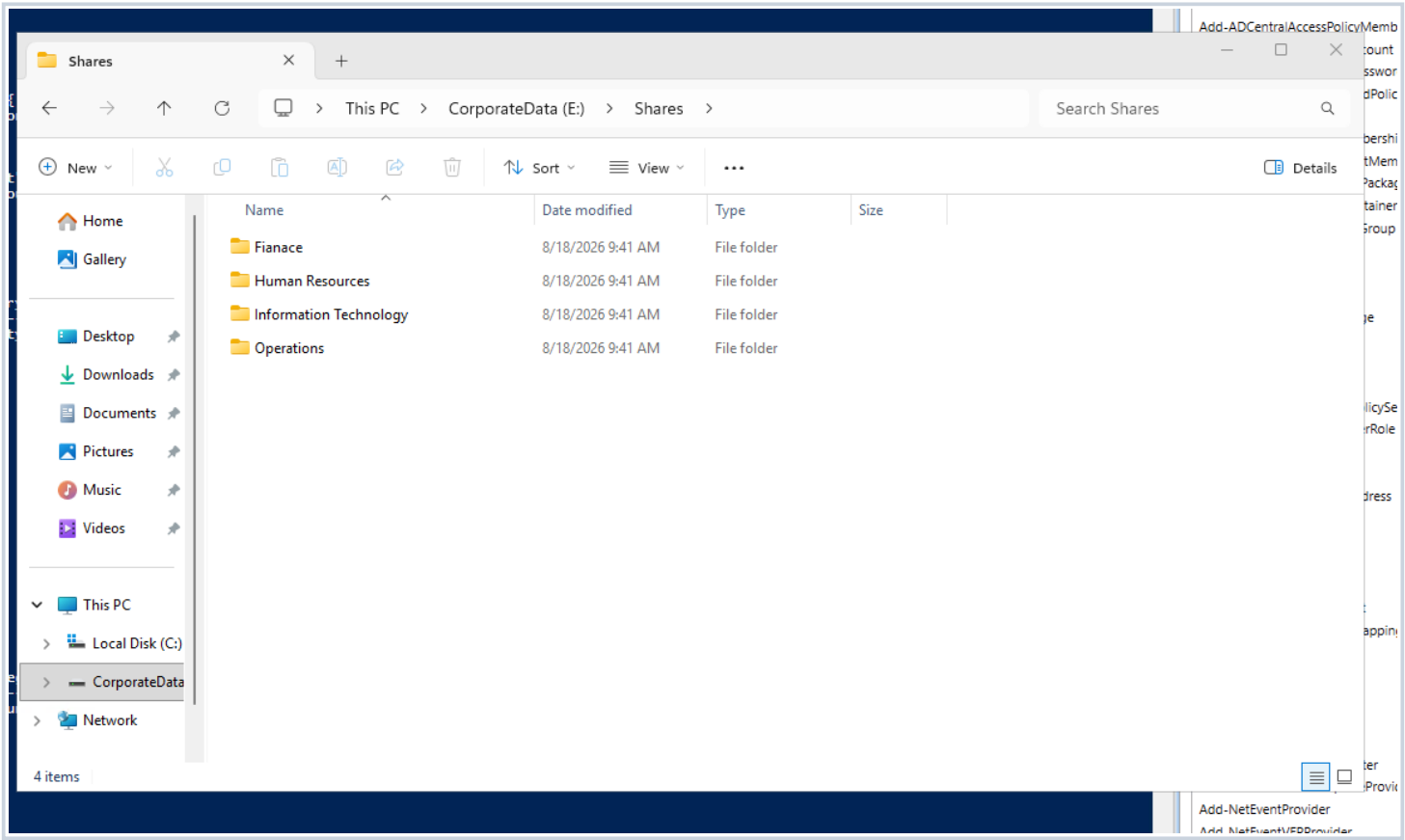


Figure 3 - File-server folder hierarchy
File Explorer confirms Finance, Human Resources, Information Technology, and Operations beneath E:\Shares.

Folder	Purpose
E:\Shares\Finance	Finance departmental data
E:\Shares\Human Resources	Human Resources departmental data
E:\Shares\Information Technology	IT departmental data
E:\Shares\Operations	Operations departmental data

STANDARD

All department folders use identical administrative controls and differ only by their matching DL_FS resource group.

GUI-Configured NTFS Permissions

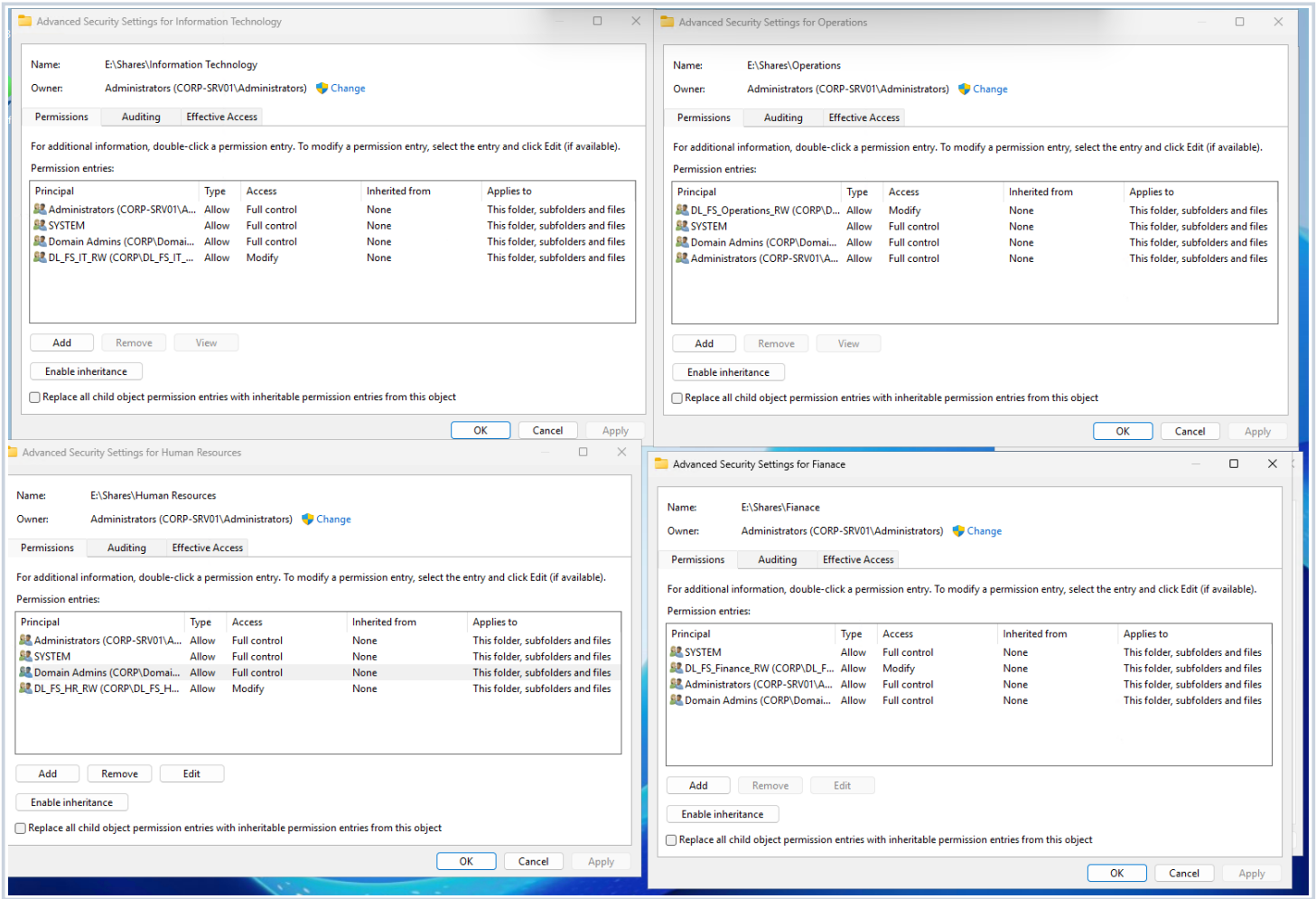


Figure 4 - Four departmental ACLs
Advanced Security Settings confirms explicit, non-inherited entries for every departmental folder.

Principal	Access	Applies to
SYSTEM	Full control	This folder, subfolders and files
CORP-SRV01\Administrators	Full control	This folder, subfolders and files
CORP\Domain Admins	Full control	This folder, subfolders and files
Matching CORP\DL_FS_*_RW	Modify	This folder, subfolders and files

PASS

Inheritance is disabled, all entries are Allow, and no individual users or unrelated departmental groups are assigned.

SMB Share Publication

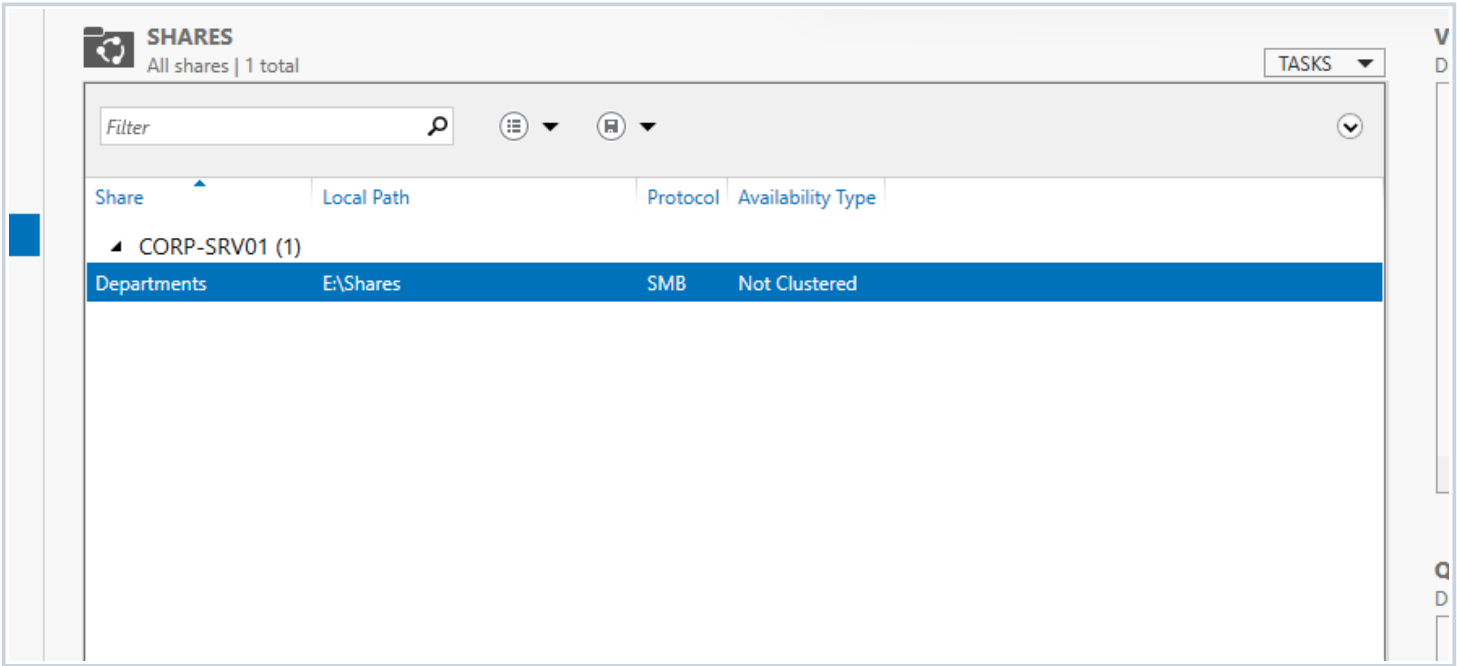


Figure 5 - Departments share in Server Manager
File and Storage Services shows Departments published from E:\Shares over SMB on CORP-SRV01.

The share presents one stable UNC path to users while NTFS ACLs divide access beneath it. This avoids maintaining four separate share definitions and allows access-based enumeration to show each user only authorized department folders.

Share-Level Design

Principal	Share right	Reason
CORP\Domain Admins	Full	Administrative management
CORP\Domain Users	Change	Connection and file operations, constrained by NTFS

SECURITY

No Everyone permission is present. NTFS departmental permissions remain the final authorization boundary.

Share Configuration Validation

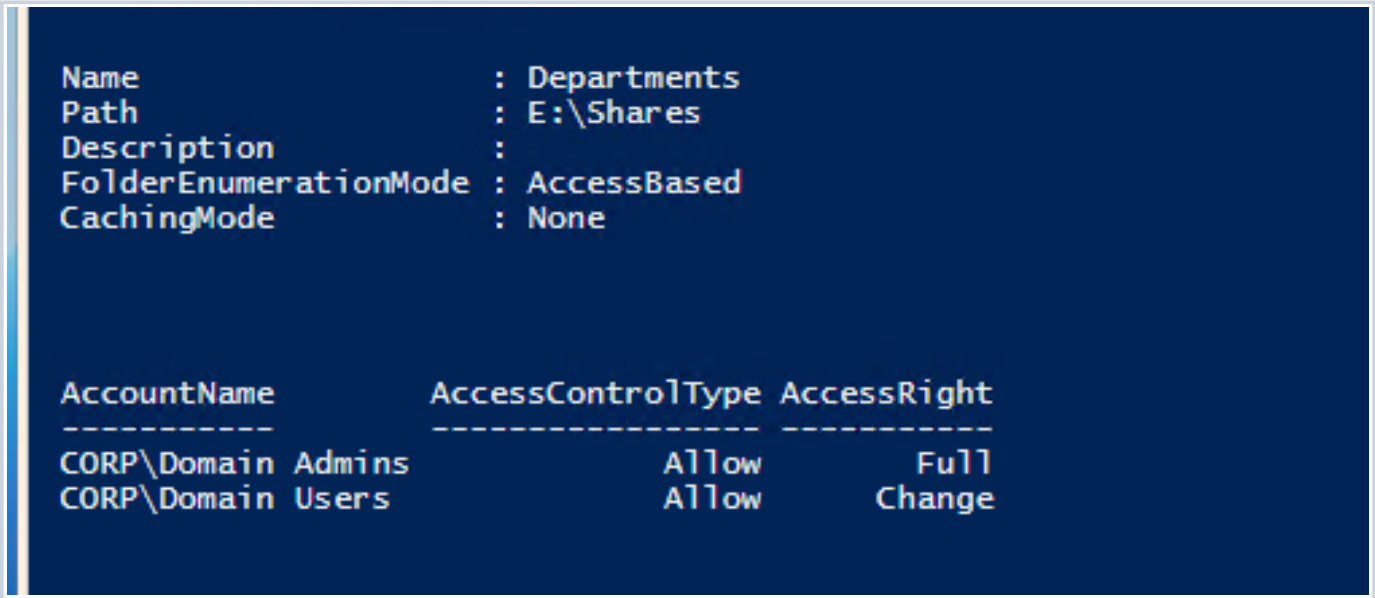


Figure 6 - PowerShell share validation
Departments maps to E:\Shares, uses AccessBased enumeration, disables caching, and carries the planned share rights.

Control	Validated value
Share name	Departments
Local path	E:\Shares
Folder enumeration	AccessBased
Caching	None
Domain Admins	Allow / Full
Domain Users	Allow / Change

PowerShell provides an authoritative configuration check independent of the creation wizard. The result verifies both availability settings and the share-level authorization layer.

PASS

The SMB definition matches the intended design with no broad Everyone entry.

Representative Effective-Access Test

```
PS C:\Users\msalazar> Get-ChildItem "\\CORP-SRV01\Departments" |
Select-Object Name,FullName |
Format-Table -AutoSize

Name                FullName
-----
Information Technology \\CORP-SRV01\Departments\Information Technology

PS C:\Users\msalazar>
PS C:\Users\msalazar> $ITTestFile =
  "\\CORP-SRV01\Departments\Information Technology\msalazar-access-test.txt"

"Authorized IT access test - $(Get-Date)" |
Set-Content -Path $ITTestFile

Get-Item $ITTestFile |
Select-Object Name,Length,
CreationTime,LastWriteTime

Name                Length CreationTime        LastWriteTime
-----
msalazar-access-test.txt    49 8/18/2026 10:35:38 AM 8/18/2026 10:35:38 AM

PS C:\Users\msalazar> "Modification successful - $(Get-Date)" |
Add-Content -Path $ITTestFile

Get-Content $ITTestFile
Authorized IT access test - 08/18/2026 10:35:38
Modification successful - 08/18/2026 10:35:56

PS C:\Users\msalazar> Get-ChildItem "\\CORP-SRV01\Departments\Finance"
Get-ChildItem : Cannot find path '\\CORP-SRV01\Departments\Finance' because it does not exist.
At line:1 char:1
+ Get-ChildItem "\\CORP-SRV01\Departments\Finance"
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (\\CORP-SRV01\Departments\Finance:String) [Get-ChildItem], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand

PS C:\Users\msalazar> New-Item `
-Path "\\CORP-SRV01\Departments\Finance\unauthorized-test.txt" `
-ItemType File
New-Item : Could not find a part of the path '\\CORP-SRV01\Departments\Finance\unauthorized-test.txt'.
At line:1 char:1
+ New-Item
+ ~~~~~
+ CategoryInfo          : WriteError: (\\CORP-SRV01\De...orized-test.txt:String) [New-Item], DirectoryNotFoundException
+ FullyQualifiedErrorId : NewItemIOError,Microsoft.PowerShell.Commands.NewItemCommand

PS C:\Users\msalazar>
```

Figure 7 - Authorized and denied access from MGMT-CL01
As msalazar, ABE exposes only Information Technology; creation and modification succeed there, while Finance lookup and file creation fail.

Test	Expected	Observed
Enumerate share root	Only Information Technology	PASS
Create IT test file	Allowed	PASS
Modify IT test file	Allowed	PASS
Browse Finance directly	Concealed or denied	PASS - PathNotFound
Create file in Finance	Concealed or denied	PASS - DirectoryNotFound

Validation Interpretation and Scope

The functional test validates the complete access chain for the standard account **CORP\msalazar**. Membership in GG_IT_Users flows through DL_FS_IT_RW to NTFS Modify on the Information Technology folder. ABE conceals Finance, and a manually entered Finance path cannot be used to browse or create content.

Why One Representative User Is Sufficient

All four departmental folders use the same ACL pattern, and the GUI evidence independently confirms the correct matching DL_FS group on each folder. Repeating the identical functional test for every department would add duplication without testing a different control. The representative test proves both the allowed and denied paths.

Scope Boundary

This project validates storage, SMB publication, NTFS authorization, ABE behavior, and one representative end-to-end access path. Quotas, file screening, shadow copies, DFS namespaces, auditing, backup, and automated drive mapping remain future projects.

Effective Permission Logic

Layer	msalazar result
Share permission	Domain Users / Change permits connection
Root visibility	Share root can be opened
IT NTFS ACL	DL_FS_IT_RW grants Modify
Finance NTFS ACL	No matching authorization
ABE presentation	Unauthorized Finance folder concealed

OUTCOME

The least-privilege control works without explicit Deny entries or direct user permissions.

Validation Matrix

Control	Status	Evidence result
Dedicated data disk	PASS	Disk 1 identified as 60-GB RAW storage
NTFS volume	PASS	E: is Healthy and labeled CorporateData
Folder hierarchy	PASS	Four departmental folders under E:\Shares
NTFS inheritance	PASS	Disabled; explicit ACLs show Inherited from None
Administrative control	PASS	SYSTEM, local Administrators, and Domain Admins have Full
Departmental control	PASS	Matching DL_FS group has Modify on each folder
SMB share	PASS	Departments published from E:\Shares
ABE and caching	PASS	AccessBased; CachingMode None
Share rights	PASS	Domain Admins Full; Domain Users Change
Authorized test	PASS	msalazar created and modified IT file
Isolation test	PASS	Finance remained concealed and unwritable

Skills Demonstrated

Windows Server file services; virtual disk provisioning; GPT/NTFS volume administration; SMB sharing; GUI-based ACL administration; AGDLP; share versus NTFS permissions; access-based enumeration; least privilege; authorized and unauthorized access testing; PowerShell validation; technical evidence capture.

COMPLETE

Project 06 completes the Permissions layer of the departmental AGDLP design established in Project 04.

Portfolio and Interview Handoff

Resume Bullet

- Built a Windows Server 2025 departmental file service using dedicated NTFS storage, SMB sharing, AGDLP-based permissions, access-based enumeration, and GUI-configured ACLs; validated authorized modification and cross-department isolation from a domain-joined Windows 11 client.

STAR Interview Story

STAR	Response
Situation	The domain had users and security groups but no centralized departmental storage or completed resource-permission layer.
Task	Create a maintainable file service that permits departmental collaboration while preventing cross-department exposure.
Action	Added a dedicated virtual disk, formatted an NTFS data volume, created four departmental folders, assigned Modify access through matching domain-local groups, published one SMB share with ABE, and tested the full allowed and denied access paths from MGMT-CL01.
Result	The authorized IT user could create and modify content while Finance remained concealed and inaccessible. The same verified ACL standard is applied consistently to all four departments.

Lessons and Production Improvements

Share permissions provide the connection boundary while NTFS ACLs enforce resource-specific authorization. ABE improves privacy and usability but does not replace permissions. A production implementation would add capacity monitoring, quotas, file screening, auditing, shadow copies, backups, high availability, and formal access reviews.

Version History

Version	Date	Change
v1.0.0	August 18, 2026	Initial completed release with storage, GUI ACL, SMB, ABE, and representative access evidence.